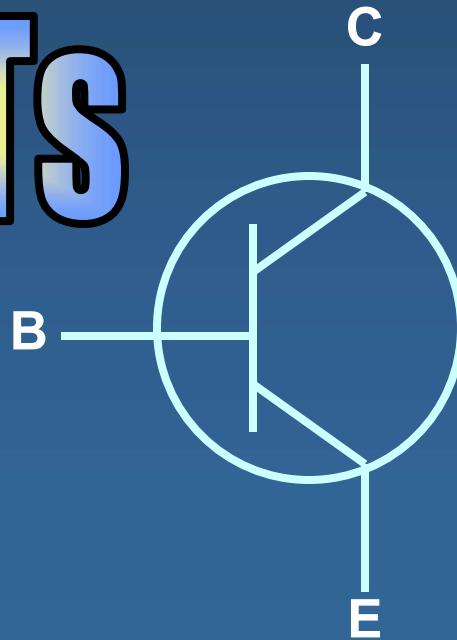


L2

Bipolar Junction Transistors

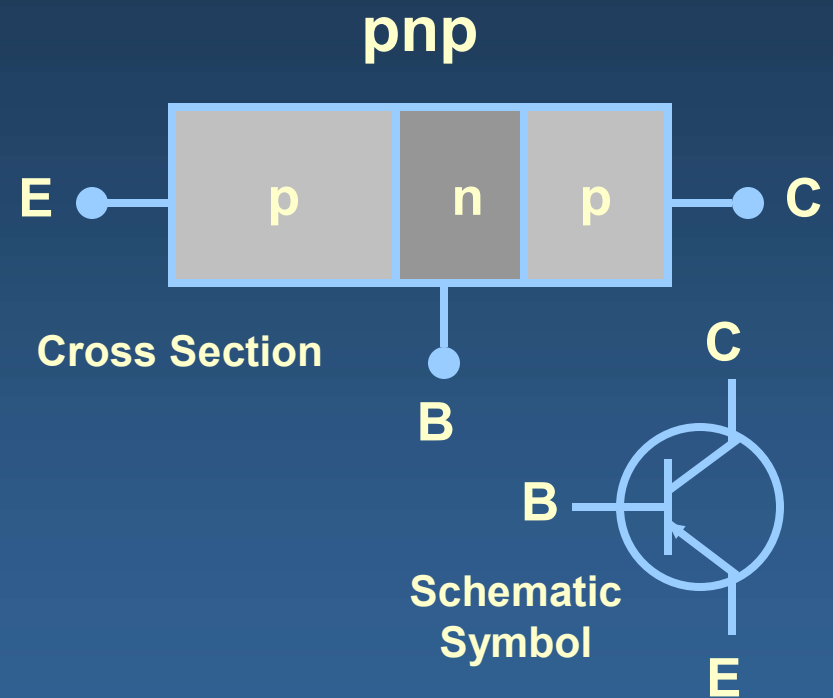
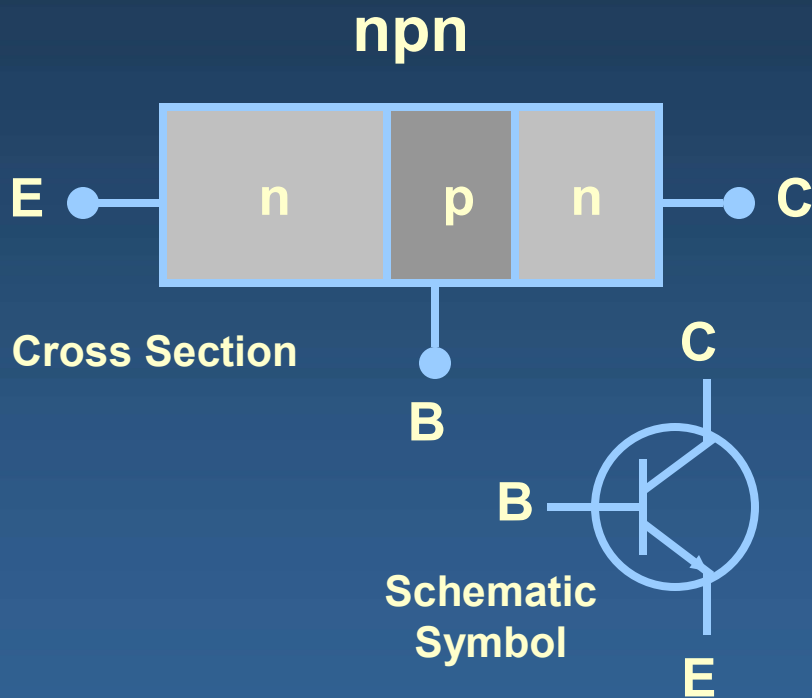
BJTs



Bipolar_Junction_Transistor (BJT)

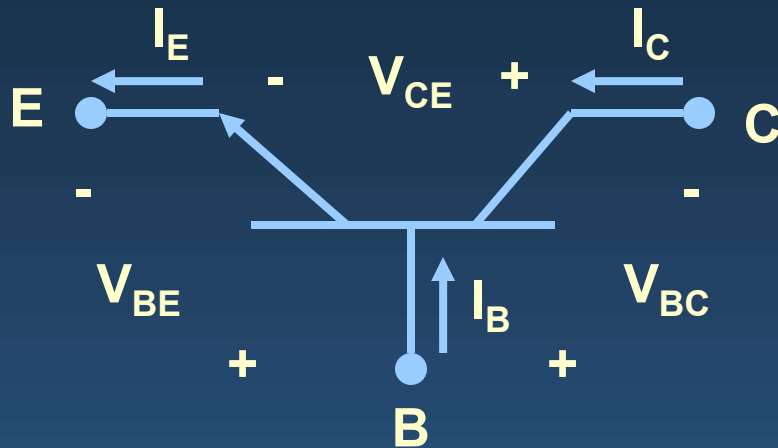
Is Constructed with three doped semiconductor regions separated by two *pn* junctions. Base emitter junction and Base collector junction .The three regions are Emitter ,Base, and Collector

The Two Types of BJT Transistors:



- Collector doping is moderately
- Base doping is slightly and very thin
- Emitter doping is heavily
- ^{NB} please know the important of each region

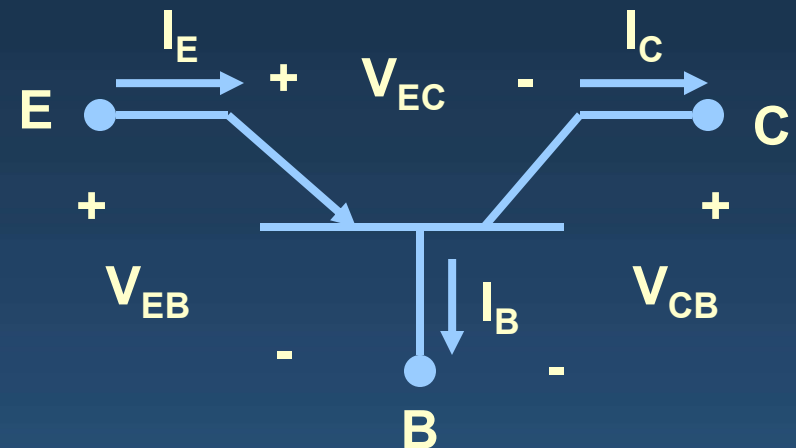
BJT_Relationships_Equations



npn

$$I_E = I_B + I_C$$

$$V_{CE} = -V_{BC} + V_{BE}$$



pnp

$$I_E = I_B + I_C$$

$$V_{EC} = V_{EB} - V_{CB}$$

Note: The equations seen above are for the transistor, not the circuit.

DC β and DC α

β = Common-emitter current gain

α = Common-base current gain

$$\beta = \frac{I_C}{I_B} \qquad \alpha = \frac{I_C}{I_E}$$

The relationships between the two parameters are:

$$\alpha = \frac{\beta}{\beta + 1}$$

$$\beta = \frac{\alpha}{1 - \alpha}$$

Note: α and β are sometimes referred to as α_{dc} and β_{dc} because the relationships being dealt with in the BJT are DC.

Relation between β and α . A simple relation exists between β and α . This follows :

$$\beta = \frac{\Delta I_C}{\Delta I_B}$$

$$\alpha = \frac{\Delta I_C}{\Delta I_E}$$

Now

$$I_E = I_B + I_C$$

or

$$\Delta I_E = \Delta I_B + \Delta I_C$$

or

$$\Delta I_B = \Delta I_E - \Delta I_C$$

Substituting the value of ΔI_B in exp. (i), we get,

$$\beta = \frac{\Delta I_C}{\Delta I_E - \Delta I_C}$$

Dividing the numerator and denominator of R.H.S. of exp. (iii) by ΔI_E , we get,

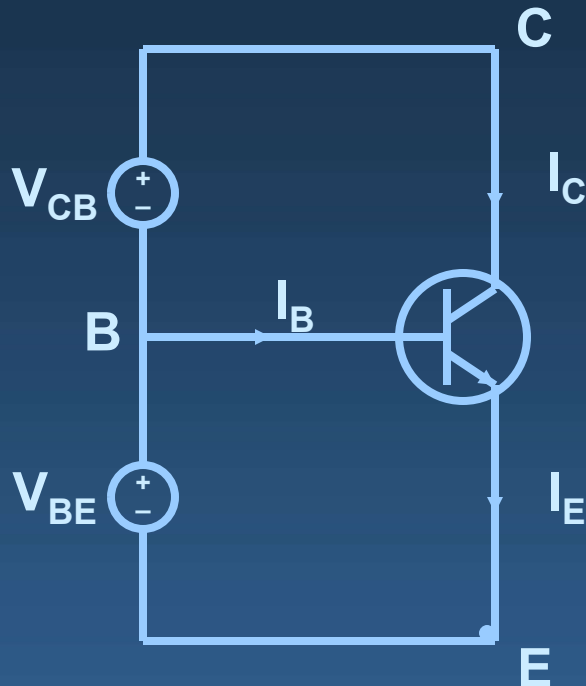
$$\beta = \frac{\frac{\Delta I_C}{\Delta I_E}}{\frac{\Delta I_E}{\Delta I_E} - \frac{\Delta I_C}{\Delta I_E}} = \frac{\alpha}{1 - \alpha}$$

$\therefore$

$$\beta = \frac{\alpha}{1 - \alpha}$$

BJT Example

Using Common-Base NPN Circuit Configuration



Given: $I_B = 50 \mu A$, $I_C = 1 \text{ mA}$

Find: I_E , β , and α

Solution:

$$I_E = I_B + I_C = 0.05 \text{ mA} + 1 \text{ mA} = 1.05 \text{ mA}$$

$$\beta = I_C / I_B = 1 \text{ mA} / 0.05 \text{ mA} = 20$$

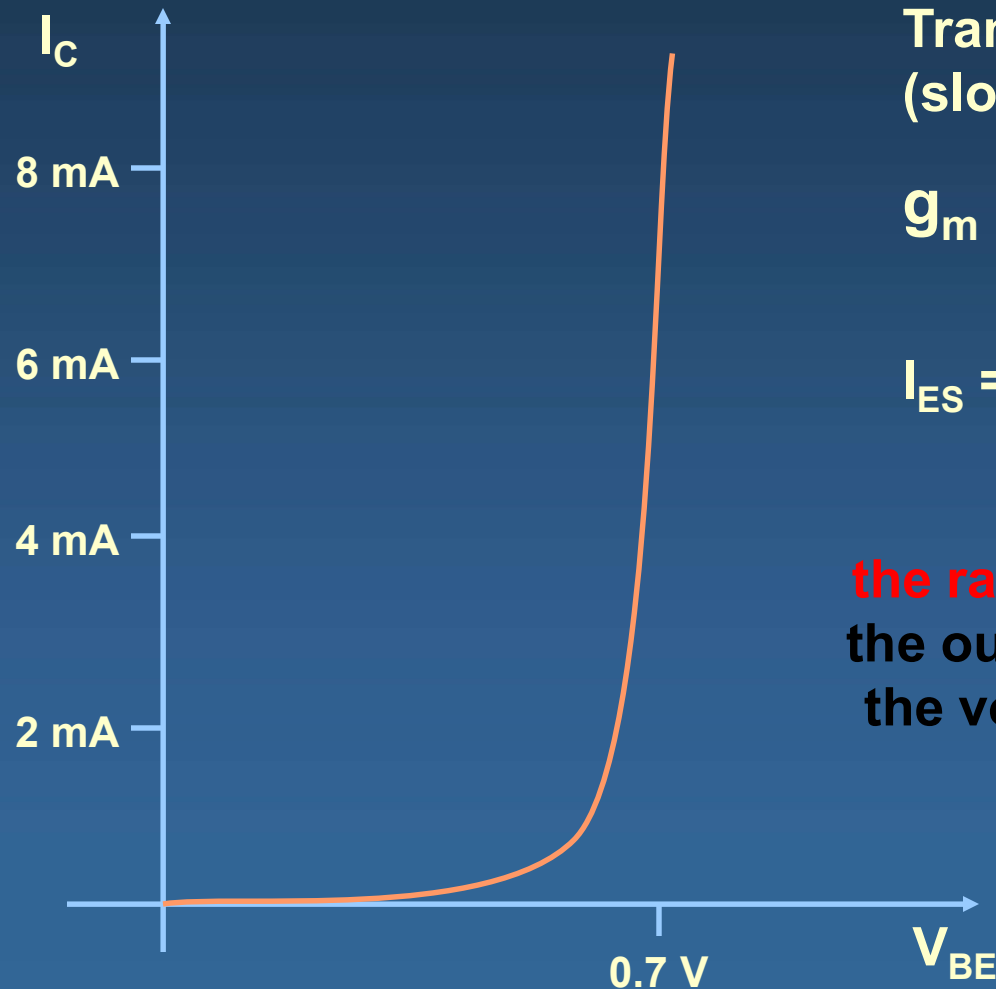
$$\alpha = I_C / I_E = 1 \text{ mA} / 1.05 \text{ mA} = 0.95238$$

α could also be calculated using the value of β with the formula from the previous slide.

$$\alpha = \frac{\beta}{\beta + 1} = \frac{20}{21} = 0.95238$$

BJT Transconductance Curve

Typical NPN Transistor



Transconductance:
(slope of the curve)

$$g_m = I_C / V_{BE}$$

I_{ES} = The reverse saturation current
of the B-E Junction.

**the ratio of the change in current at
the output terminal to the change in
the voltage at the input terminal of
an active device**

Modes of Operation

Active:

- Most important mode of operation
- Central to amplifier operation
- The region where current curves are practically flat

Saturation:

- Barrier potential of the junctions cancel each other out causing a virtual short short (closed switch)

Cutoff:

- Current reduced to zero
- Ideal transistor behaves like an (open switch)

* Note: There is also a mode of operation called inverse active, but it is rarely used.

Three Types of BJT Biasing

Biasing the transistor refers to applying voltage to get the transistor to achieve certain operating conditions.

Common-Base Biasing (CB) :	input = V_{EB} & I_E output = V_{CB} & I_C
Common-Emitter Biasing (CE):	input = V_{BE} & I_B output = V_{CE} & I_C
Common-Collector Biasing (CC):	input = V_{BC} & I_B output = V_{EC} & I_E

Common Base Connection (biasing)

In this circuit arrangement, input is applied between emitter and base and output is taken from collector

and base. Here, base of the transistor is common to both input and output circuits and hence the

name common base connection. In Fig. 8.9 (i), a common base npn transistor circuit is shown whereas

Fig. 8.9 (ii) shows the common base pnp transistor circuit.

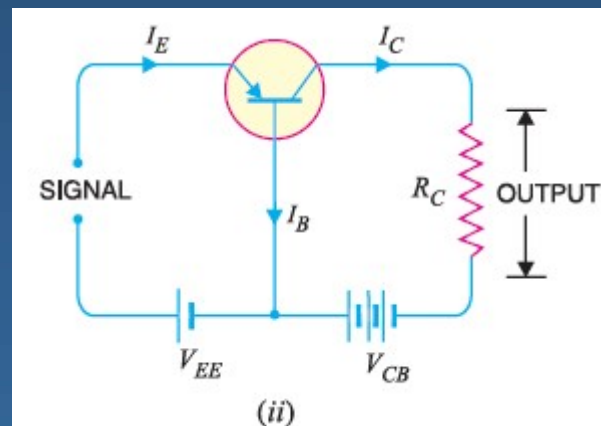
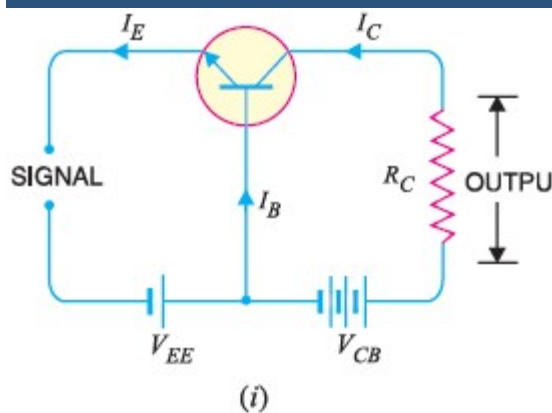


Fig. 8.9

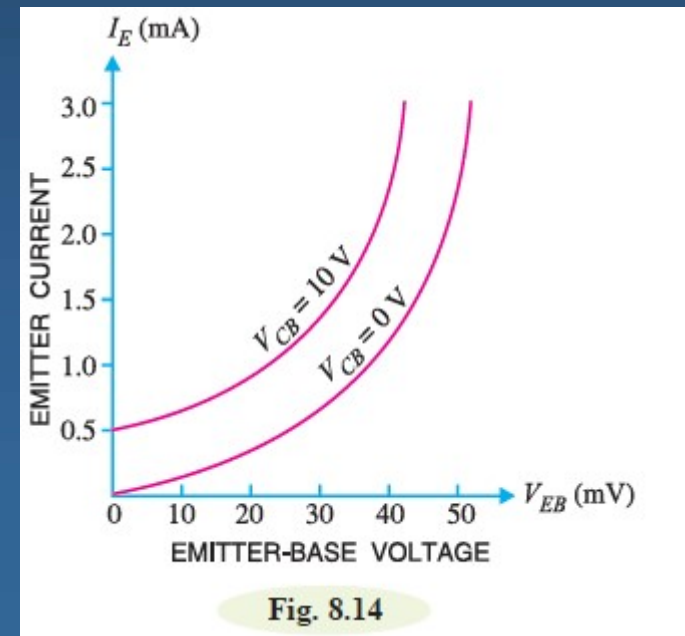
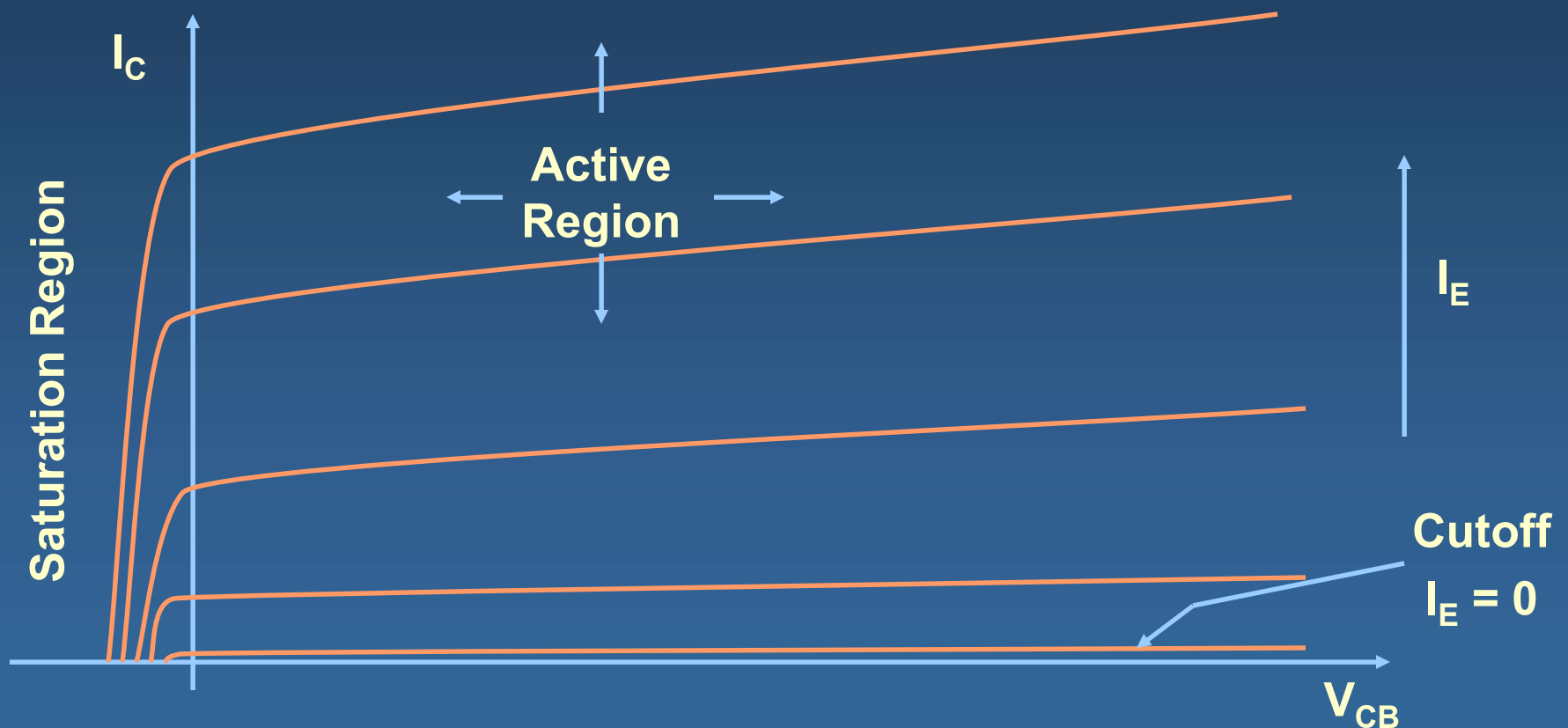


Fig. 8.14

Common-Base_Output_Characteristic

Although the Common-Base configuration is not the most common biasing type, it is often helpful in the understanding of how the BJT works.

Emitter-Current Curves



Common Emitter Connection

In this circuit arrangement, input is applied between base and emitter and output is taken from the

collector and emitter. Here, emitter of the transistor is common to both input and output circuits and

hence the name common emitter connection.

Fig. 8.16 (i) shows common emitter npn transistor

circuit whereas Fig. 8.16 (ii) shows common emitter pnp transistor circuit.

I_E has to be kept constant because any change in I_E will produce corresponding change in I_C .

Here, we are

interested to see how V_{CB} influences I_C .

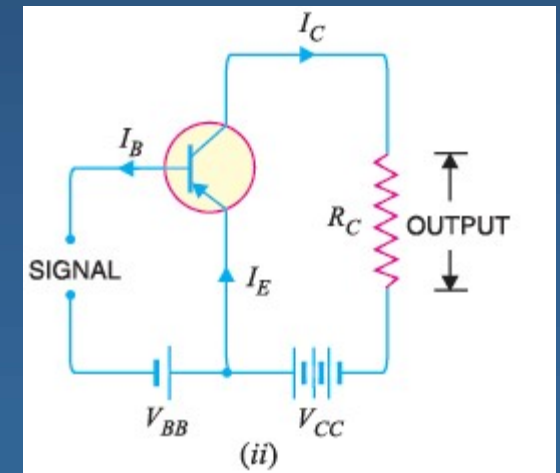
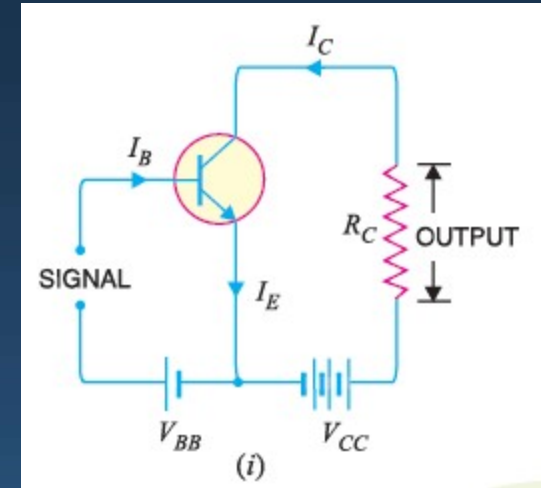
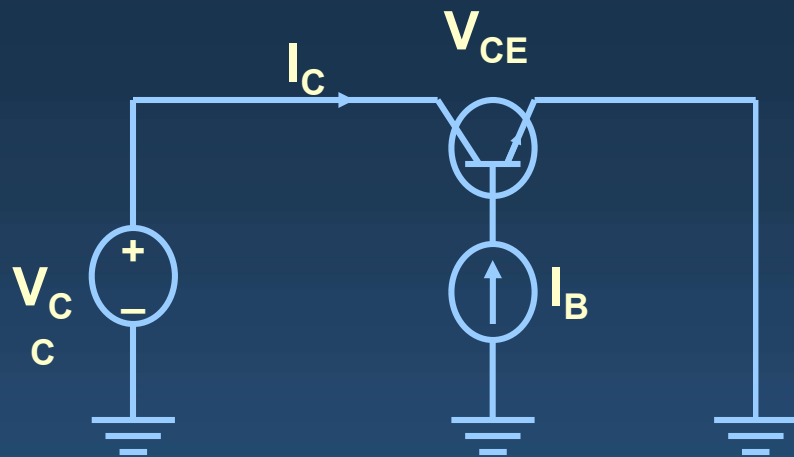


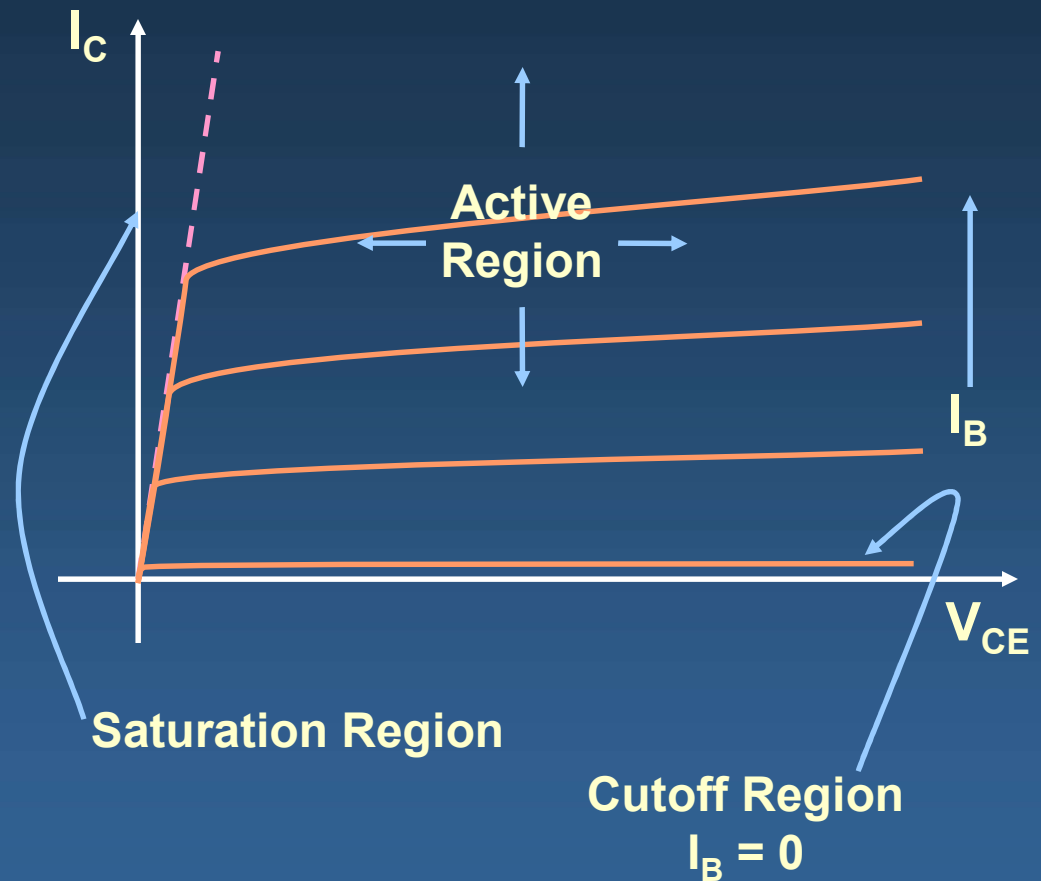
Fig. 8.16

Common-Emitter_output_Characteristic

Circuit Diagram



Collector-Current Curves



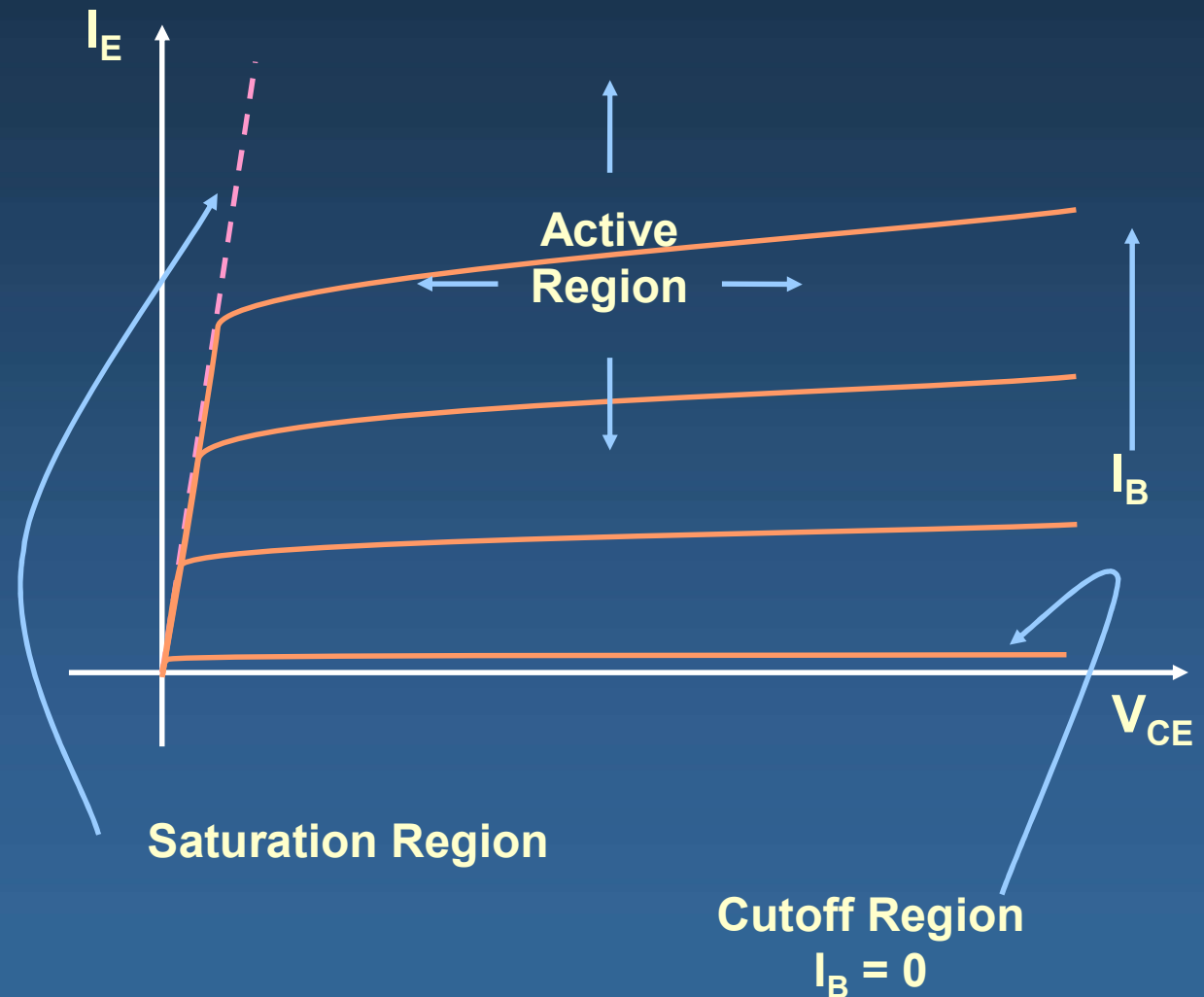
Region of Operation	Description
Active	Small base current controls a large collector current
Saturation	$V_{CE(sat)} \sim 0.2V$, V_{CE} increases with I_C
Cutoff	Achieved by reducing I_B to 0, Ideally, I_C will also equal 0.

Common-Collector

Emitter-Current Curves

The Common-Collector biasing circuit is basically equivalent to the common-emitter biased circuit except instead of looking at I_C as a function of V_{CE} and I_B we are looking at I_E .

Also, since $\alpha \sim 1$, and $\alpha = I_C/I_E$ that means $I_C \sim I_E$



Transistor Circuit as an Amplifier

A transistor raises the strength of a weak signal and thus acts as an amplifier. Fig. 8.7 shows the basic circuit of a transistor amplifier. The weak signal is applied between emitter-base junction and output is taken across the load R_C connected in the collector circuit. In order to achieve faithful amplification, the input circuit should always remain forward biased. To do so, a d.c. voltage V_{EE} is applied in the input circuit in addition to the signal as

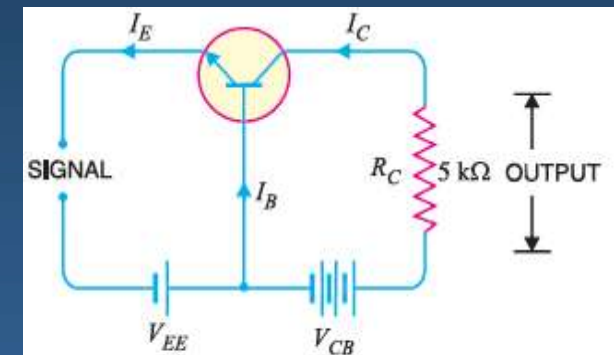


Fig. 8.7

shown. This d.c. voltage is known as bias voltage and its magnitude is such that it always keeps the input circuit forward biased regardless of the polarity of the signal.

As the input circuit has low resistance, therefore, a small change in signal voltage causes an appreciable change in emitter current. This causes almost the *same change in collector current due to transistor action. The collector current flowing through a high load resistance *RC produces a large* voltage across it. Thus, a weak signal applied in the input circuit appears in the amplified form in the collector circuit. It is in this way that a transistor acts as an amplifier.

THE BJT AS AN AMPLIFIER

Amplification is the process of linearly **increasing the amplitude of an electrical signal** and is one of the major properties of a transistor. As you learned, a BJT exhibits current gain (called β). When a BJT is biased in the active (or linear) region, as previously described, the BE junction **has a low resistance due to forward bias** and the BC junction **has a high resistance due to reverse bias**.

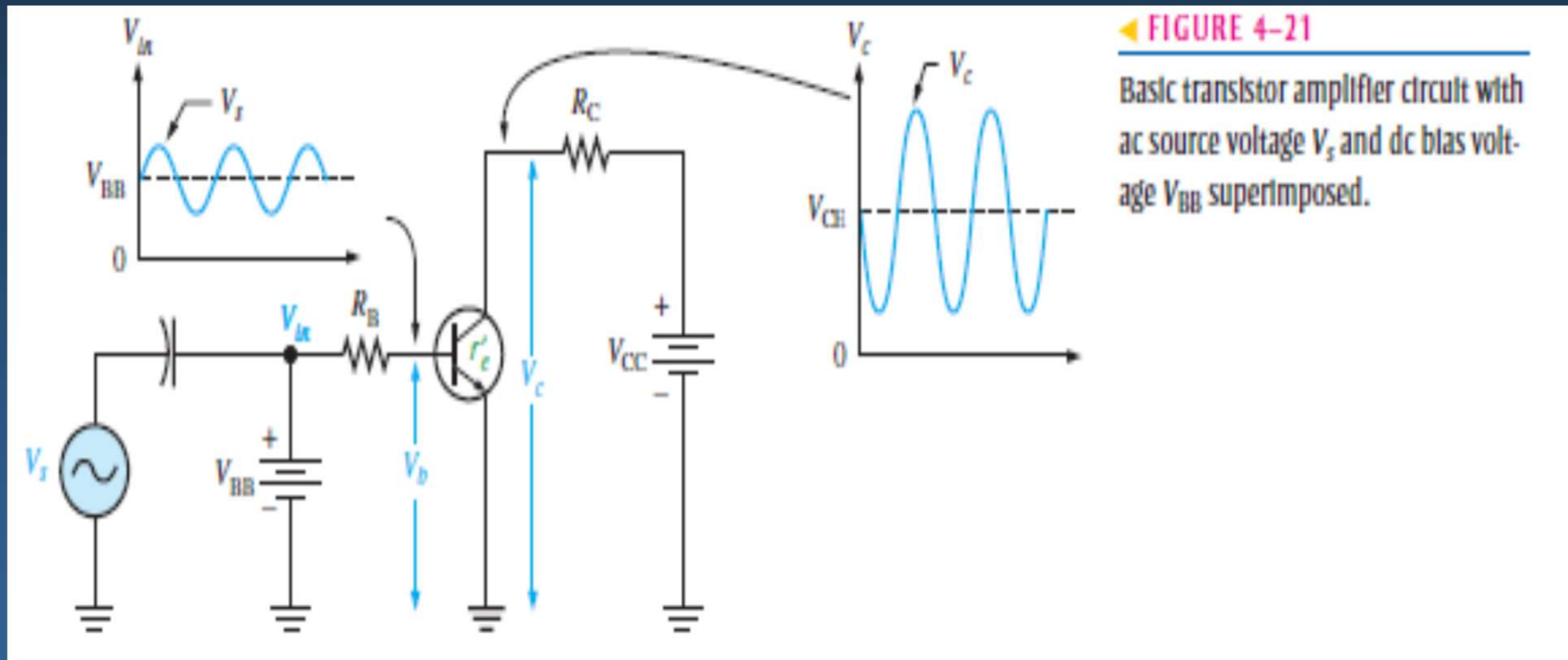
Voltage Amplification

- As you have learned, a transistor amplifies current because the collector current is equal to the base current multiplied by the current gain β .

$I_c = I_b \times \beta$ The base current in a transistor is very small compared to the collector and emitter currents. Because of this, the collector current is approximately equal to the emitter current. ie $I_c = I_e$

- With this in mind, let's look at the circuit in Figure 4–21. An ac voltage, V_s , is superimposed on the dc bias voltage V_{BB} by capacitive coupling as shown. The dc bias voltage V_{CC} is connected to the collector through the collector resistor, R_C .

Voltage Amplification



- The ac input voltage produces an ac base current, which results in a much larger ac collector current. The ac collector current produces an ac voltage across RC , thus producing an amplified, but inverted, reproduction of the ac input voltage in the active region of operation, as illustrated in Figure 4–21.
- The forward-biased base-emitter junction presents a very low resistance to the ac signal. This internal ac emitter resistance is designated r'_e in Figure 4–21 and appears in series with RB

The ac base voltage is

$$V_b = I_e r'_e$$

The ac collector voltage, V_c , equals the ac voltage drop across R_C .

$$V_c = I_c R_C$$

Since $I_c \cong I_e$, the ac collector voltage is

$$V_c \cong I_e R_C$$

V_b can be considered the transistor ac input voltage where $V_b = V_s - I_b R_B$. V_c can be considered the transistor ac output voltage. Since *voltage gain* is defined as the ratio of the output voltage to the input voltage, the ratio of V_c to V_b is the ac voltage gain, A_v , of the transistor.

$$A_v = \frac{V_c}{V_b}$$

Substituting $I_e R_C$ for V_c and $I_e r'_e$ for V_b yields

$$A_v = \frac{V_c}{V_b} \cong \frac{I_e R_C}{I_e r'_e}$$

The I_e terms cancel; therefore,

$$A_v \cong \frac{R_C}{r'_e}$$

Equation 4–7

Equation 4–7 shows that the transistor in Figure 4–21 provides amplification in the form of voltage gain, which is dependent on the values of R_C and r'_e .

THE BJT AS A SWITCH

- In the previous section, you saw how a BJT can be used as a linear amplifier. The second major application area is switching applications. When used as an electronic switch, a BJT is normally operated alternately in cutoff and saturation. Many digital circuits use the BJT as a switch.

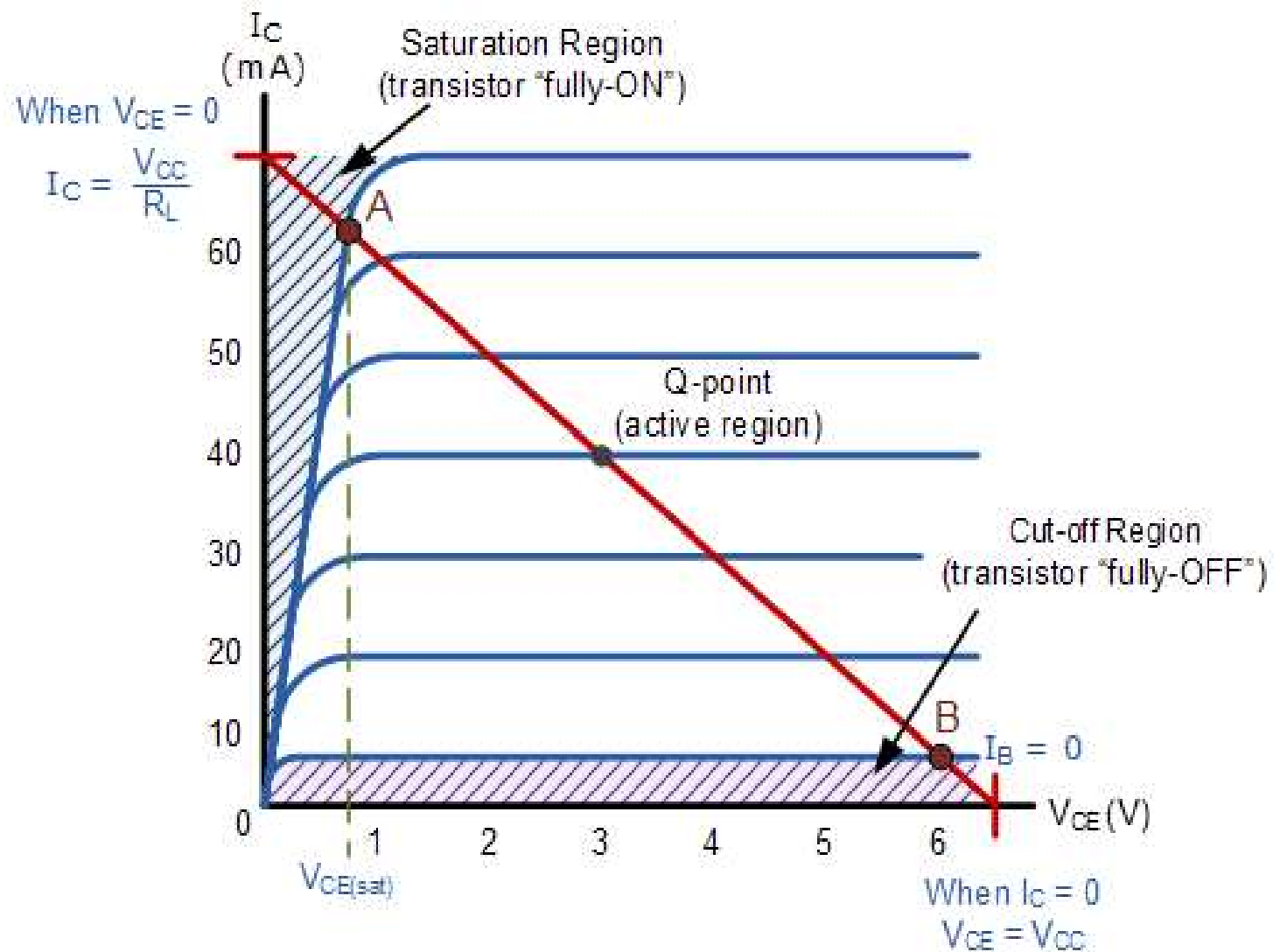
Cutoff=open switch =high resistance =off

Saturation =closed switch =low resistance = on

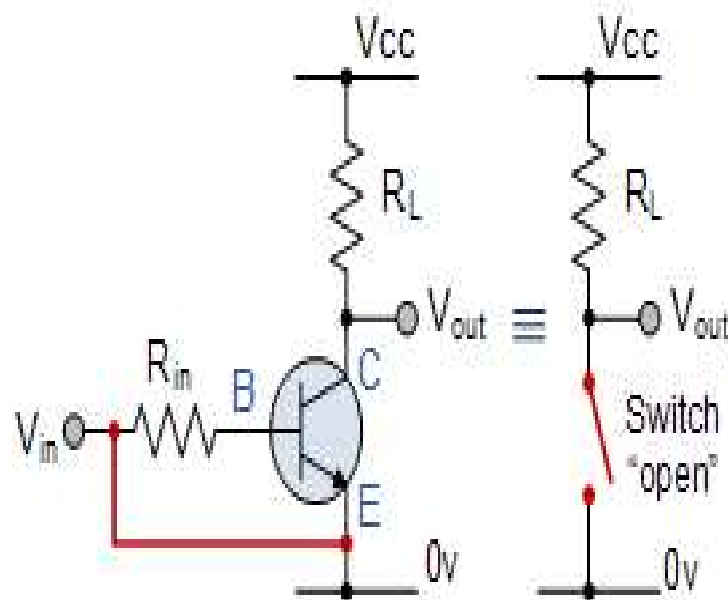
Switching Operation

- Figure 4–23 illustrates the basic operation of a BJT as a switching device. In part (a), the transistor is in the cutoff region because the base-emitter junction is not forward-biased. In this condition, there is, ideally, an *open* between collector and emitter, as indicated by the switch equivalent. In part (b), the transistor is in the saturation region because the base emitter junction and the base-collector junction are forward-biased and the base current is made large enough to cause the **collector current to reach its saturation value. In this condition, there is, ideally, a *short* between collector and emitter,** as indicated by the switch equivalent. Actually, a small voltage drop across the transistor of up to a few tenths of a volt normally occurs, which is the saturation voltage, $V_{CE(sat)}$.

Operating Regions

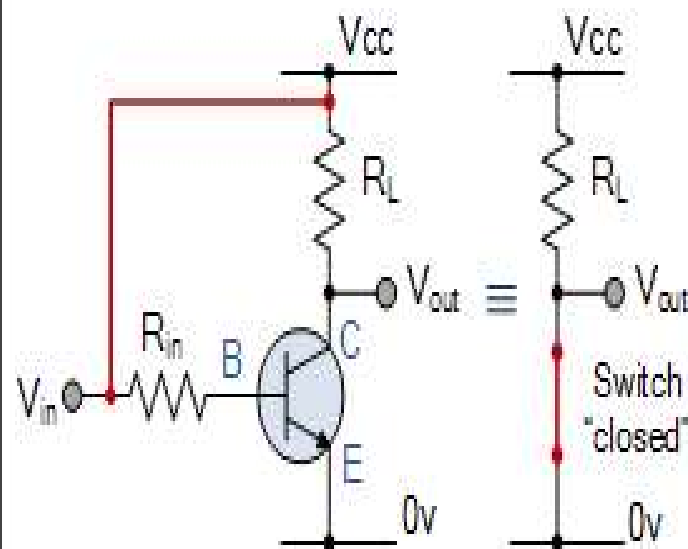


Cut-off Characteristics



- The input and Base are grounded (0V)
- Base-Emitter voltage $V_{BE} < 0.7V$
- Base-Emitter junction is reverse biased
- Base-Collector junction is reverse biased
- Transistor is "fully-OFF" (Cut-off region)
- No Collector current flows ($I_C = 0$)
- $V_{OUT} = V_{CE} = V_{CC} = "1"$
- Transistor operates as an "open switch"

Saturation Characteristics



- The input and Base are connected to V_{CC}
- Base-Emitter voltage $V_{BE} > 0.7V$
- Base-Emitter junction is forward biased
- Base-Collector junction is forward biased
- Transistor is "fully-ON" (saturation region)
- Max Collector current flows ($I_C = V_{CC}/R_L$)
- $V_{CE} = 0$ (ideal saturation)
- $V_{OUT} = V_{CE} = "0"$
- Transistor operates as a "closed switch"